

How Good Is Your Synthetic Social Data? Conditional Alignment and Novelty Across 34 GSS Waves

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Abstract. Synthetic populations and simulated survey respondents are increasingly used in computational social science, but marginal and task-based validation can miss the conditional dependencies that organize social attitudes. We introduce a model-agnostic framework based on full conditional relationships. A MAP-alignment score compares the probability assigned to each observed response with the modal conditional probability under the same respondent context; the complete normalized conditional profile separately induces a metric on strictly positive finite-state generative processes. We pair MAP alignment with nearest-real row similarity to measure respondent-level copying. Across 34 General Social Survey waves from 1972–2024, spanning 197–1,273 modeled variables per wave, a recursive conditional Large Science Model (LSM) achieved mean MAP alignment 0.8032, compared with 0.8004 for the original (control), for retention 1.0034 (wave-bootstrap 95% confidence interval $[0.9996, 1.0079]$). Relative to the independent-marginal null, the excess MAP-alignment fraction was 1.062 ($[0.992, 1.154]$). Mean nearest-real similarity was 0.3800, corresponding to row novelty 0.6200, whereas the original (control) had essentially zero novelty. The framework separates conditional alignment from respondent-level reuse and provides a validation tool for synthetic social data.

Keywords: synthetic populations · computational social science · survey simulation · General Social Survey · conditional inference · model validation

1 Introduction and Motivation

Synthetic social data support data sharing, replication, and population simulation, but their validity remains difficult to assess. This problem is increasingly consequential as generative systems are used to construct synthetic human samples and survey respondents [1,3]. Records may appear plausible individually while failing to reproduce empirical response dependencies [20]. A useful validation criterion should therefore test population structure without rewarding copies of observed respondents.

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Current practices fall broadly into:

- **Likelihood-based metrics** (e.g., ELBO [19] and perplexity [17]), requiring the generator to have a tractable joint density;
- **Task-based metrics** [11], such as training a classifier on synthetic data and testing on real data, which depend on subjective downstream tasks and can obscure structural mismatches;
- **Embedding-based metrics** such as FID [13] or classifier-based two-sample tests [21], which rely on feature extractors unrelated to the data domain and limit interpretability.

Comparisons of componentwise moments or covariance matrices are also insufficient [25,23]: they capture pairwise linear structure but can miss nonlinear and higher-order dependencies. Section 7 gives two distributions with identical first- and second-order moments but different conditional structure.

The recursive conditional architecture used by the LSM comparator has supported digital twins of the infant microbiome [24] and influenza evolution [28]. In this benchmark, each wave-specific LSM models survey variables through coordinate-wise conditional models and generates complete records by recursive conditional sampling, following the same general architecture. Here we study evaluation rather than substantive opinion dynamics, using repeated social surveys as the test bed.

We evaluate conditional structure inferred directly from data. Given real or synthetic samples, we estimate each coordinate’s conditional distribution $\phi^i(\cdot \mid x^{-i})$, for example using conditional inference trees, and define the MAP-alignment score

$$v(x, i) = \frac{\phi^i(x^i \mid x^{-i})}{\max_y \phi^i(y \mid x^{-i})}, \quad (1)$$

For each coordinate, the numerator evaluates the observed value under the real-data conditional and the denominator is its modal probability in the same context. Averaging $v(x, i)$ over records and coordinates yields the bounded statistic $\Upsilon(D)$. It measures graded agreement with conditional maximum-a-posteriori predictions [4] and requires only samples and a conditional learner.

Under positivity and compatibility, a complete system of full conditionals identifies the joint distribution [5,8,12,2]. This motivates a one-sided MAP-alignment comparison and a symmetric distance between full conditional profiles. Unlike aggregate utility measures [27,9,10,7,18], we assess conditional structure and proximity to individual real records separately.

2 Formal Setup for Conditional Analysis

Let

$$X = (X^1, \dots, X^N) \quad (2)$$

take values in the finite product space

$$\mathcal{X} = \mathcal{X}^1 \times \dots \times \mathcal{X}^N. \quad (3)$$

Let P be a strictly positive data-generating distribution. For coordinate i , define the full conditional

$$P_i(x^i \mid x^{-i}) := P(X^i = x^i \mid X^{-i} = x^{-i}). \quad (4)$$

A learned conditional kernel is denoted by $\phi^i(\cdot \mid x^{-i})$. In practice, a small ε -floor may be added and the conditional vector renormalized. Positivity is used together with compatibility; it does not by itself guarantee that an arbitrary family of kernels corresponds to a joint distribution.

3 MAP-Alignment Functional

For any model $\{\phi^i\}$ and sample $x \in \mathcal{X}$, define

$$v(x, i) := \frac{\phi^i(x^i \mid x^{-i})}{\max_{y \in \mathcal{X}^i} \phi^i(y \mid x^{-i})}, \quad (5)$$

which equals 1 exactly when x^i is a maximizer of the model conditional.

Given a dataset $D = \{x_k\}_{k=1}^M$, define

$$\Upsilon(D) := \frac{1}{MN} \sum_{k=1}^M \sum_{i=1}^N v(x_k, i), \quad (6)$$

an empirical estimate of

$$\Upsilon_\phi(P) = \mathbb{E}_{X \sim P} \left[\frac{1}{N} \sum_{i=1}^N v(X, i) \right]. \quad (7)$$

3.1 Behavior Under Exact Conditionals

Lemma 1 (MAP-Alignment Under Exact Conditionals). *Assume $\phi^i = P_i$ for all i . Fix i and x^{-i} , and let $p_j := P_i(j \mid x^{-i})$, $p_{\max} := \max_j p_j$. If $X^i \sim P_i(\cdot \mid x^{-i})$, then*

$$v(X, i) = \frac{P_i(X^i \mid x^{-i})}{p_{\max}}, \quad \mathbb{E}[v(X, i) \mid x^{-i}] = \frac{1}{p_{\max}} \sum_j p_j^2. \quad (8)$$

Moreover:

- $v(X, i) = 1$ iff $X^i \in \arg \max_j p_j$.
- If $P_i(\cdot \mid x^{-i})$ is uniform on its support, then $\mathbb{E}[v(X, i) \mid x^{-i}] = 1$.
- If some $p_j < p_{\max}$, then $\mathbb{E}[v(X, i) \mid x^{-i}] < 1$.

Proof. Immediate from the definition and the fact that $\sum_j p_j^2 \leq p_{\max} \sum_j p_j = p_{\max}$.

Thus, under exact conditionals, $v(X, i)$ is a normalized conditional-likelihood score. Its population target is

$$\Upsilon_{\text{oracle}}(P) := \mathbb{E}_{X \sim P} \left[\frac{1}{N} \sum_{i=1}^N \frac{P_i(X^i \mid X^{-i})}{\max_y P_i(y \mid X^{-i})} \right]. \quad (9)$$

This target is generally smaller than one when the exact conditional is nonuniform. Consequently, synthetic MAP alignment is assessed by agreement with a real-data reference score, not by requiring $\mathcal{T} \rightarrow 1$.

4 Brook–Dobrushin Factorization and Identifiability

Brook’s lemma [5] and related consistency results [8] show how a compatible system of full conditionals identifies a strictly positive joint distribution.

Assume

$$\phi^i(x^i \mid x^{-i}) = P_i(x^i \mid x^{-i}) \quad \text{whenever } P(x) > 0. \quad (10)$$

Choose a reference configuration x° with $P(x^\circ) > 0$. Define the interpolating sequence

$$x^{(0)} = x^\circ, \quad x^{(i)} = (x^1, \dots, x^i, x^{i+1,\circ}, \dots, x^{N,\circ}), \quad (11)$$

so $x^{(N)} = x$. By repeated conditioning,

$$\frac{P(x^{(i)})}{P(x^{(i-1)})} = \frac{P_i(x^i \mid x^{<i}, x^{>i,\circ})}{P_i(x^{i,\circ} \mid x^{<i}, x^{>i,\circ})}. \quad (12)$$

Multiplying yields the Brook factorization [5]:

$$\frac{P(x)}{P(x^\circ)} = \prod_{i=1}^N \frac{P_i(x^i \mid x^{<i}, x^{>i,\circ})}{P_i(x^{i,\circ} \mid x^{<i}, x^{>i,\circ})}. \quad (13)$$

Replacing P_i by ϕ^i on the support gives an explicit reconstruction of P from the conditionals.

Theorem 1 (Identification from compatible full conditionals). *Let P and Q be strictly positive distributions on the same finite product space. If their full conditionals agree for every coordinate and every configuration, then $P = Q$.*

Proof. Brook’s factorization reconstructs each joint probability ratio relative to a fixed reference configuration from the full conditional system. Equality of all full conditionals therefore gives identical probability ratios under P and Q ; normalization gives $P = Q$ [5,8]. The compatibility assumption is essential: positivity alone does not guarantee that an arbitrary collection of conditional kernels corresponds to a joint law.

5 Evaluating Generators via $\mathcal{T}$

For any model $\{\phi^i\}$,

$$\mathcal{T}_\phi(P) = \mathbb{E}_{X \sim P} \left[\frac{1}{N} \sum_{i=1}^N \frac{\phi^i(X^i \mid X^{-i})}{\max_y \phi^i(y \mid X^{-i})} \right]. \quad (14)$$

Let D_{test} be i.i.d. data from P . If $\phi_n^i \rightarrow P_i$ pointwise on the support and satisfy strict positivity, then dominated convergence yields

$$\mathcal{T}_{\phi_n}(P) \rightarrow \mathcal{T}_{\text{oracle}}(P). \quad (15)$$

Corollary 1 (Conditional-Kernel Convergence and MAP-Alignment Consistency). *Let ϕ_n^i be strictly positive kernels converging to P_i on $\text{supp}(P)$. Let $\tilde{P}_n$ denote the joint compatible with $\{\phi_n^i\}$. Then:*

- $\tilde{P}_n \rightarrow P$ pointwise (equivalently, in total variation on the finite state space).
- For any sequence of test sets D_{test} with $|D_{\text{test}}| \rightarrow \infty$, the empirical scores $\Upsilon(D_{\text{test}}; \phi_n)$ converge in probability to $\Upsilon_{\text{oracle}}(P)$.

Both conclusions follow from the assumed convergence of the full conditional kernels. Convergence, or numerical agreement, of the scalar MAP-alignment score alone does not imply convergence of the conditional kernels or recovery of the joint law. Systematically low MAP alignment can nevertheless reveal disagreement with the evaluator’s conditional modes even when marginal summaries look satisfactory.

6 Conditional Inference, MAP-Alignment Uncertainty, and a Metric on Underlying Processes

6.1 Inference of Conditionals Using Conditional Learners

Let $D = \{x^k\}_{k=1}^n$ be a training dataset in a finite product space. We construct a family of full conditional models $\Phi^{(n)} = \{\varphi^{i,n}(\cdot | x^{-i})\}_{i=1}^N$ by solving one supervised prediction problem for each coordinate. In the experiments, these models are implemented using conditional inference trees, whose unbiased permutation-based split selection is described in [16,15,26].

The theory does not require a tree-specific convergence exponent. We instead state the estimation assumption explicitly: for each coordinate,

$$\begin{aligned} \sup_{x^{-i}, x^i} |\varphi^{i,n}(x^i | x^{-i}) - P_i(x^i | x^{-i})| &\leq \delta_{i,n}, \\ \delta_n := \max_i \delta_{i,n} &\xrightarrow{P} 0. \end{aligned} \tag{16}$$

A small ε -floor may be applied before renormalization to maintain strict positivity on the effective support. When the learned kernels are compatible, the resulting full conditional system identifies a unique joint law by Theorem 1.

6.2 MAP-Alignment and Its Finite-Sample Uncertainty

Given a trained conditional family $\Phi^{(n)}$, define

$$v_{\Phi^{(n)}}(x, i) = \frac{\varphi^{i,n}(x^i | x^{-i})}{\max_{y \in \mathcal{X}^i} \varphi^{i,n}(y | x^{-i})} \in [0, 1]. \tag{17}$$

For an independent test dataset $D_{\text{test}} = \{x_k\}_{k=1}^M$, let

$$Z_k := \frac{1}{N} \sum_{i=1}^N v_{\Phi^{(n)}}(x_k, i), \quad \hat{\Upsilon}_{\Phi^{(n)}}(D_{\text{test}}) := \frac{1}{M} \sum_{k=1}^M Z_k. \tag{18}$$

The coordinates within a record may be dependent, but the row scores Z_k are independent and bounded in $[0, 1]$ under i.i.d. test sampling. Hence Hoeffding's inequality [14] gives

$$\Pr\left(\left|\hat{\mathcal{Y}}_{\Phi^{(n)}} - \mathbb{E}Z_1\right| > \epsilon\right) \leq 2 \exp(-2M\epsilon^2). \quad (19)$$

Because each finite conditional vector has modal probability at least $1/|\mathcal{X}^i|$, the normalization in Eq. (17) is Lipschitz in the conditional vector. Combining test-sample variation with Eq. (16) yields

$$\hat{\mathcal{Y}}_{\Phi^{(n)}}(D_{\text{test}}) = \mathcal{Y}_{\text{oracle}}(P) + O_p\left(M^{-1/2} + \delta_n\right). \quad (20)$$

6.3 A Metric on Underlying Generative Processes

For a strictly positive process P with full conditionals P_i , define its population v -profile:

$$u_P(x, i) = \frac{P_i(x^i \mid x^{-i})}{\max_{y \in \mathcal{X}^i} P_i(y \mid x^{-i})}. \quad (21)$$

Fix a reference probability measure μ on $\mathcal{X}$ with full support, and sample the coordinate index uniformly from $\{1, \dots, N\}$. Define the distance

$$d_\mu(P_1, P_2) = \mathbb{E}_{X \sim \mu} \left[\frac{1}{N} \sum_{i=1}^N |u_{P_1}(X, i) - u_{P_2}(X, i)| \right]. \quad (22)$$

Theorem 2. *Let μ have full support on the finite product space $\mathcal{X}$. Then d_μ is a metric on the set of strictly positive probability distributions on $\mathcal{X}$.*

Proof. Nonnegativity and symmetry are immediate. The triangle inequality follows from the scalar inequality $|a - c| \leq |a - b| + |b - c|$ applied pointwise and averaged over coordinates and integrated with respect to μ . For identity of indiscernibles, if $d_\mu(P_1, P_2) = 0$, then $u_{P_1}(x, i) = u_{P_2}(x, i)$ for every (x, i) because $\mathcal{X}$ is finite and μ has full support. For each fixed (x^{-i}, i) , the conditional distribution is recovered from its normalized profile by $P_i(x^i \mid x^{-i}) = u_P(x, i) / \sum_{z \in \mathcal{X}^i} u_P((z, x^{-i}), i)$. Thus all full conditionals agree. Theorem 1 then implies $P_1 = P_2$.

Given empirical datasets D_1, D_2 , train conditional generator families G_1, G_2 yielding conditional families $\Phi^{(1)}$ and $\Phi^{(2)}$. Independently draw a common evaluation sample $E_\mu = \{x^{(k)}\}_{k=1}^M$ i.i.d. from the fixed reference measure μ . Define the empirical distance

$$\hat{d}_\mu(D_1, D_2; E_\mu) = \frac{1}{MN} \sum_{k=1}^M \sum_{i=1}^N \left| u_{G_1}(x^{(k)}, i) - u_{G_2}(x^{(k)}, i) \right|. \quad (23)$$

For notational convenience, let $d_\mu(G_1, G_2)$ denote Eq. (22) evaluated using the two learned normalized profiles.

Theorem 3. Assume the conditional estimators satisfy (16) with uniform errors ϵ_1, ϵ_2 . Then

$$|\hat{d}_\mu(D_1, D_2; E_\mu) - d_\mu(P_1, P_2)| = O_p(M^{-1/2} + \epsilon_1 + \epsilon_2). \quad (24)$$

Proof. By the triangle inequality,

$$|\hat{d}_\mu - d_\mu(P_1, P_2)| \leq |\hat{d}_\mu - d_\mu(G_1, G_2)| \quad (25)$$

$$+ |d_\mu(G_1, G_2) - d_\mu(P_1, P_2)|. \quad (26)$$

For the first term, the row-level quantities

$$Z_k = \frac{1}{N} \sum_{i=1}^N |u_{G_1}(x^{(k)}, i) - u_{G_2}(x^{(k)}, i)| \quad (27)$$

are i.i.d. and bounded in $[0, 1]$ because E_μ is independent of the training datasets. Hoeffding's inequality therefore gives the $O_p(M^{-1/2})$ term. The second term is controlled by the Lipschitz continuity of $u(\varphi) = \varphi / \max_y \varphi(y)$ on finite probability simplices, together with (16).

6.4 One-Sided MAP-Alignment Comparison Against Held-Out Real Data

Let D_{fit} be an evaluator-training dataset independent of both the real reference dataset D_{ref} and the synthetic dataset D_{syn} . Fit a conditional family $\hat{\Phi}^{(\text{real})}$ on D_{fit} , and evaluate both the real reference records and synthetic records under that same fixed conditional family:

$$\mathcal{Y}_{\text{ref}} = \hat{\mathcal{Y}}_{\hat{\Phi}^{(\text{real})}}(D_{\text{ref}}), \quad \mathcal{Y}_{\text{syn}} = \hat{\mathcal{Y}}_{\hat{\Phi}^{(\text{real})}}(D_{\text{syn}}). \quad (28)$$

We report the signed MAP-alignment gap and retention ratio

$$\Delta\mathcal{Y} := \mathcal{Y}_{\text{syn}} - \mathcal{Y}_{\text{ref}}, \quad R_{\mathcal{Y}} := \frac{\mathcal{Y}_{\text{syn}}}{\mathcal{Y}_{\text{ref}}}. \quad (29)$$

A generator matching the reference in average MAP alignment should have $\Delta\mathcal{Y}$ near zero, or equivalently $R_{\mathcal{Y}}$ near one. Negative gaps indicate lower average MAP alignment; positive gaps do not by themselves establish superiority because finite-sample estimation and mode concentration can increase the normalized score. When an independent evaluator-training dataset is unavailable, cross-fitting may instead be used to avoid evaluating real records under a model trained on those same records.

6.5 Consistency of the Dataset Distance as a Metric on Processes

Theorem 4 (Convergence to the Population Metric). Let D_1 and D_2 be drawn i.i.d. from strictly positive processes P_1 and P_2 , and let E_μ be an independent i.i.d. sample from the fixed full-support reference measure μ . Suppose the learned conditional kernels $\hat{P}_{k,i}$ satisfy

$$\max_{1 \leq i \leq N} \sup_{x^{-i}, x^i} |\hat{P}_{k,i}(x^i | x^{-i}) - P_{k,i}(x^i | x^{-i})| \quad (30)$$

$$\leq \epsilon_k, \quad k \in \{1, 2\}. \quad (31)$$

Algorithm 1: Cross-fitted conditional MAP alignment and row novelty

Input : Real data D_{real} ; synthetic data D_{syn} ; K folds; conditional learner Learn ; level α .
Output: $\hat{\gamma}_{\text{ref}}$, $\hat{\gamma}_{\text{syn}}$, $\Delta\hat{\gamma}$, $R_{\mathcal{I}}$, and row novelty.

- 1 Partition D_{real} into folds $F_1, \dots, F_K$
- 2 **for** $k \leftarrow 1$ **to** K **do**
- 3 Fit $\Phi_k \leftarrow \text{Learn}(D_{\text{real}} \setminus F_k)$
- 4 Score each $x \in F_k$ using Φ_k and store its row-average MAP alignment
- 5 Score an assigned subset of D_{syn} using the same Φ_k
- 6 Average the stored real and synthetic row scores to obtain $\hat{\gamma}_{\text{ref}}$ and $\hat{\gamma}_{\text{syn}}$
- 7 Set $\Delta\hat{\gamma} \leftarrow \hat{\gamma}_{\text{syn}} - \hat{\gamma}_{\text{ref}}$ and $R_{\mathcal{I}} \leftarrow \hat{\gamma}_{\text{syn}}/\hat{\gamma}_{\text{ref}}$
- 8 For each synthetic row, compute nearest-real similarity η_j and report row novelty $1 - m^{-1} \sum_j \eta_j$
- 9 Use a paired bootstrap over independent datasets or survey waves for the final confidence interval
- 10 **return** all summary statistics

with $\epsilon_k \rightarrow 0$ in probability as $|D_k| \rightarrow \infty$, and suppose $M = |E_\mu| \rightarrow \infty$. Then

$$\hat{d}_\mu(D_1, D_2; E_\mu) \xrightarrow{p} d_\mu(P_1, P_2). \quad (32)$$

Proof. From the previous theorem we have

$$|\hat{d}_\mu(D_1, D_2; E_\mu) - d_\mu(P_1, P_2)| = O_p\left(M^{-1/2} + \epsilon_1 + \epsilon_2\right). \quad (33)$$

As $M \rightarrow \infty$ and $|D_k| \rightarrow \infty$, both $M^{-1/2}$ and ϵ_k converge to zero in probability, which implies (32).

Corollary 2. If $P_1 = P_2$, then

$$\hat{d}_\mu(D_1, D_2; E_\mu) \xrightarrow{p} 0. \quad (34)$$

If $P_1 \neq P_2$, then

$$\hat{d}_\mu(D_1, D_2; E_\mu) \xrightarrow{p} d_\mu(P_1, P_2) > 0. \quad (35)$$

6.6 Cross-Fitted Evaluation When Independent Training Data Are Unavailable

Algorithm 1 gives a general cross-fitted implementation for settings in which a separate evaluator-training dataset is unavailable. In the GSS benchmark reported here, the wave-specific conditional evaluators were trained on data disjoint from the corresponding real reference records and from all generated samples. The reported GSS scores are therefore out-of-sample with respect to evaluator fitting and do not require the cross-fitting procedure in Algorithm 1.

For a fixed fitted conditional family and M independent evaluated rows, the row-level Hoeffding radius at confidence $1 - \delta$ is

$$r_M(\delta) = \sqrt{\frac{1}{2M} \log \frac{2}{\delta}}. \quad (36)$$

At $\delta = 0.05$, the radii are 0.1358 for $M = 100$ and 0.0429 for $M = 1000$. Conditional-estimation uncertainty must be handled separately through data splitting, repeated fitting, or a higher-level bootstrap.

7 Low-Order Moment Matching is Insufficient: Two Illustrative Examples

Two examples in $\mathbb{R}^3$ show that identical first- and second-order moments do not imply similar conditional structure. We use the continuous-density analogue of the finite-state score, replacing the maximum by an essential supremum. A formal extension to continuous and mixed-type data is left for future work.

7.1 Example 1: Uniform vs. Gaussian with Identical Moments

Consider the following two distributions on $\mathbb{R}^3$:

$$X^{(U)} \sim \text{Uniform}([- \sqrt{3}, \sqrt{3}]^3), \quad X^{(G)} \sim \mathcal{N}_3(0, I_3). \quad (37)$$

Both satisfy

$$\mathbb{E}[X^{(U)}] = \mathbb{E}[X^{(G)}] = 0, \quad \text{Cov}(X^{(U)}) = \text{Cov}(X^{(G)}) = I_3. \quad (38)$$

Hence all marginal means, variances, and the full covariance matrix coincide.

Yet their conditional structures differ sharply. For $X^{(U)}$, each coordinate has a flat conditional density on $[-\sqrt{3}, \sqrt{3}]$, yielding

$$v(X^{(U)}, i) = 1 \quad \text{for almost every sample.} \quad (39)$$

For $X^{(G)}$, the i th coordinate conditional is $X_i^{(G)} | X_{-i}^{(G)} \sim \mathcal{N}(0, 1)$, giving

$$v(X^{(G)}, i) = \exp\left(-\frac{1}{2}X_i^{(G)2}\right), \quad (40)$$

with mean exactly $1/\sqrt{2} \approx 0.7071$. Thus, although the datasets match in mean and covariance, the MAP-alignment values differ substantially:

$$\gamma(X^{(U)}) = 1, \quad \gamma(X^{(G)}) = \frac{1}{\sqrt{2}}. \quad (41)$$

Low-order moments fail to detect this discrepancy.

7.2 Example 2: Matching Non-Identity Covariances with Distinct Conditional Structure

To demonstrate that the limitation persists even when the covariance matrix is nontrivial, apply the same invertible linear map

$$L = \begin{pmatrix} 1 & \rho & 0 \\ 0 & 1 & \rho \\ 0 & 0 & 1 \end{pmatrix}, \quad 0 < \rho < 1, \quad (42)$$

to both datasets and define

$$Y^{(U)} = LX^{(U)}, \quad Y^{(G)} = LX^{(G)}. \quad (43)$$

Both transformed datasets have the *same* non-identity covariance matrix

$$\text{Cov}(Y^{(U)}) = \text{Cov}(Y^{(G)}) = \Sigma = LL^\top, \quad (44)$$

and share identical columnwise means and variances.

Their conditional distributions nevertheless differ. Since $Y^{(U)}$ is the image of a uniform cube under a linear shear, each conditional $Y^{i,(U)} \mid Y^{-i,(U)}$ is uniform on a finite interval (given by the intersection of a line with a parallelepiped), implying

$$v(Y^{(U)}, i) = 1 \quad \text{for almost every sample.} \quad (45)$$

Conversely, $Y^{(G)} \sim \mathcal{N}_3(0, \Sigma)$ has linear-Gaussian conditionals. If $Y^{i,(G)} \mid Y^{-i,(G)} \sim \mathcal{N}(m_i(y^{-i}), \sigma_i^2)$, then

$$v(Y^{(G)}, i) = \exp\left(-\frac{1}{2}(y^i - m_i(y^{-i}))^2 / \sigma_i^2\right), \quad (46)$$

and the standardized conditional residual is $\mathcal{N}(0, 1)$, so its mean is exactly $1/\sqrt{2} \approx 0.7071$.

Thus, even though $Y^{(U)}$ and $Y^{(G)}$ agree in all first- and second-order statistics, their conditional behavior differs markedly, and the MAP-alignment statistic again reveals the discrepancy:

$$\Upsilon(Y^{(U)}) = 1, \quad \Upsilon(Y^{(G)}) = \frac{1}{\sqrt{2}}. \quad (47)$$

7.3 Implications

Thus, matching means and covariance does not determine conditional structure. MAP alignment detects the discrepancy in both examples.

8 Applications: Comparing Synthesizers

8.1 Row-Level Memorization Diagnostic

MAP alignment does not detect row reuse: a dataset can score highly while copying observed records. We therefore add a row-level novelty diagnostic.

Let $D_{\text{real}} = \{x^{(1)}, \dots, x^{(n)}\}$ and $D_{\text{syn}} = \{y^{(1)}, \dots, y^{(m)}\}$ be real and synthetic datasets defined over the same categorical (or discretized) feature space. Using a concatenated one-hot embedding $f(\cdot)$, we define cosine similarity

$$s(u, v) = \frac{\langle f(u), f(v) \rangle}{\|f(u)\|_2 \|f(v)\|_2}, \quad (48)$$

which, for categorical data, reduces to the fraction of coordinates on which two records agree.

For each synthetic record $y^{(j)}$, we define its nearest-real similarity

$$\eta_j = \max_{1 \leq i \leq n} s(x^{(i)}, y^{(j)}). \quad (49)$$

We summarize row-level reuse via the *lack-of-novelty score*

$$\mathcal{N}(D_{\text{syn}}) = \frac{1}{m} \sum_{j=1}^m \eta_j, \quad (50)$$

with larger values indicating lower novelty. In particular, $\mathcal{N} = 1$ corresponds to exact row reuse (e.g., bootstrap resampling), while smaller values indicate increasing deviation from any individual real record. Tail statistics of $\{\eta_j\}$ (e.g., the 95th percentile) capture worst-case near-copying behavior.

The diagnostics are complementary: $\mathcal{Y}(D)$ measures conditional alignment, whereas $\mathcal{N}(D_{\text{syn}})$ measures proximity to specific real records. Nearest-row novelty is a memorization diagnostic, not a formal privacy guarantee; disclosure risk requires separate membership, linkage, or attribute-inference analysis. The target is $\Delta\mathcal{Y} \rightarrow 0$ (equivalently $R_{\mathcal{Y}} \rightarrow 1$) with $\mathcal{N}$ well below one.

8.2 Practical Use of the Diagnostics

The diagnostics answer different questions and should be selected accordingly. MAP alignment is appropriate when the primary concern is whether synthetic records preserve the conditional response structure represented by a held-out real-data evaluator. In that setting, practitioners should report the signed MAP-alignment gap, the raw reference score, and, when an independent-marginal null is available, the zero-anchored excess-alignment fraction A .

Nearest-real similarity is appropriate when the concern is copying or near reuse of individual records. It should be reported together with exact duplicate rates and upper-tail summaries, such as the 95th or 99th percentile, rather than only a mean. It is not a substitute for formal disclosure-risk analysis.

Neither diagnostic replaces marginal checks or application-specific validation. When the synthetic data will support a particular estimand, classifier, or policy simulation, MAP alignment and novelty should be combined with relevant marginal, task-based, and, where necessary, privacy metrics. A single scalar score is not sufficient for all uses of synthetic social data.

9 Empirical Evaluation on Repeated GSS Cross-Sections

9.1 Benchmark Design

We analyzed 34 General Social Survey (GSS) waves spanning 1972–2024 [22]. Each wave was split before model fitting: 50% of respondents were used to train the wave-specific LSM and conditional evaluator, and the remaining 50% were held out from LSM fitting for comparator construction, the real-data control, and all reported evaluation. Missing and nonresponse entries were preserved as missing throughout LSM fitting and evaluation rather than recoded as substantive response levels or statistically imputed. For nearest-row similarity, missing entries were represented by a common missing marker solely for one-hot computation. Across waves, the data contained 71,667 respondent records and 197–1,273 modeled variables per wave (Table 1).

We compared LSM, a restricted Chow–Liu hybrid [6], CTGAN [29], an independent marginal baseline, and the original (control). Each condition produced 1,000 records and was scored by the same fixed, wave-specific evaluator. Waves

Table 1. GSS data characteristics and analysis split.

Characteristic	Value
Analyzed waves	34
Survey years covered	1972–2024
Respondent records	71,667
Respondents per wave	1,372–4,510
LSM/evaluator training split	50% per wave
Held-out analysis split	50% per wave
Modeled variables per wave	197–1,273
Evaluated records per condition	1,000

Table 2. Cross-wave benchmark summary over 34 GSS waves. The signed gap is $\mathcal{Y}_{\text{generator}} - \mathcal{Y}_{\text{control}}$. The excess-alignment fraction Λ is defined in Eq. (51). Confidence intervals are nonparametric 95% bootstrap intervals over waves; the null and control entries in the Λ column are fixed anchors. Row novelty is one minus mean nearest-real similarity.

Generator	Mean $\mathcal{Y}$	Signed gap (95% CI)	Excess Λ (95% CI)	Nearest-real similarity	Row novelty
LSM	0.8032	0.0027 [−0.0004, 0.0062]	1.062 [0.992, 1.154]	0.3800	0.6200
Chow–Liu hybrid	0.7872	−0.0132 [−0.0159, −0.0106]	0.698 [0.661, 0.736]	0.6560	0.3440
CTGAN	0.7630	−0.0375 [−0.0415, −0.0332]	0.143 [0.068, 0.216]	0.5665	0.4335
Independent baseline	0.7567	−0.0437 [−0.0483, −0.0390]	0.000 (anchor)	0.5619	0.4381
Original (control)	0.8004	0.0000	1.000 (anchor)	1.0000	0.0000

were the paired comparison units. We report the signed gap $\mathcal{Y}_{g,w} - \mathcal{Y}_{\text{control},w}$, the retention ratio $\mathcal{Y}_{g,w}/\mathcal{Y}_{\text{control},w}$, and row novelty. Confidence intervals are nonparametric bootstraps over the 34 waves. Because GSS waves are temporally ordered repeated cross-sections rather than exchangeable draws from a population of survey years, these intervals summarize stability across the observed waves and are not population-level temporal confidence intervals. Numerical similarity values were clipped to $[0, 1]$.

Because $\mathcal{Y}$ is bounded but not zero-anchored, we additionally report the fraction of alignment recovered above the independent-marginal null. Let $\bar{\mathcal{Y}}_g$ denote the mean MAP-alignment score of generator g across waves. We define

$$\Lambda_g = \frac{\bar{\mathcal{Y}}_g - \bar{\mathcal{Y}}_{\text{ind}}}{\bar{\mathcal{Y}}_{\text{control}} - \bar{\mathcal{Y}}_{\text{ind}}}. \quad (51)$$

Thus, the independent-marginal baseline is anchored at $\Lambda = 0$ and the original (control) at $\Lambda = 1$. Bootstrap intervals resample waves jointly and recompute the complete ratio within each replicate. Table 2 summarizes the cross-wave results.

9.2 MAP Alignment

Across the 34 waves, LSM attained mean $\mathcal{Y} = 0.8032$ versus 0.8004 for the original (control). The signed difference was 0.0027 (wave-bootstrap 95% CI [−0.0004, 0.0062]), and retention was 1.0034 (95% CI [0.9996, 1.0079]). LSM exceeded the control in 22 waves, but the interval includes parity; we therefore

interpret this as control-level MAP alignment, not superiority or equality of the full conditional distributions.

The restricted Chow–Liu hybrid retained 0.9834 of control alignment (95% CI [0.9801, 0.9866]), CTGAN retained 0.9530 (95% CI [0.9479, 0.9584]), and the independent baseline retained 0.9451 (95% CI [0.9394, 0.9510]). These ratios are not zero-anchored. On the excess-alignment scale in Eq. (51), LSM attained $A_{\text{LSM}} = 1.062$ (95% CI [0.992, 1.154]), the restricted Chow–Liu hybrid attained 0.698 ([0.661, 0.736]), and CTGAN attained 0.143 ([0.068, 0.216]). The LSM interval includes one, so its point estimate above one indicates consistency with control-level alignment, not superiority to real data. Figure 1(b) shows the paired wave-level signed-gap distributions, while Fig. 1(d) shows MAP-alignment retention across GSS waves.

9.3 MAP Alignment Without Row Reuse

The original (control) had nearest-real similarity essentially one. Mean similarity was 0.3800 for LSM, 0.6560 for the restricted Chow–Liu hybrid, 0.5665 for CTGAN, and 0.5619 for the independent baseline, corresponding to novelty 0.6200, 0.3440, 0.4335, and 0.4381, respectively. Figure 1(c) shows the wave-level novelty distributions; Fig. 1(a) combines mean novelty with excess MAP alignment above the independent-marginal null. LSM therefore occupies the favorable region of the archived alignment–novelty plane.

10 Discussion

Synthetic-data quality has two distinct axes. The original (control) preserves MAP alignment but provides no novelty, whereas independent sampling creates new rows while degrading conditional alignment. The desired regime combines reference-level MAP alignment with separation from observed respondents.

The GSS benchmark spans surveys with different dimensionality and conditional structure. Raw retention ratios are useful for within-wave comparison but are not zero-anchored; the excess-alignment scale therefore measures recovery above the dependence-destroying independent-marginal null. On this scale, LSM is consistent with the original (control), whereas the other archived configurations recover smaller fractions of the control-to-null increment. The nearest-row diagnostic excludes simple resampling as the explanation. For computational social science, the framework provides a model-agnostic validation layer for synthetic survey populations used in sharing, replication, and simulation.

11 Limitations

The identification results require strict positivity and compatibility of the full conditional system. Separately fitted conditional learners need not be exactly

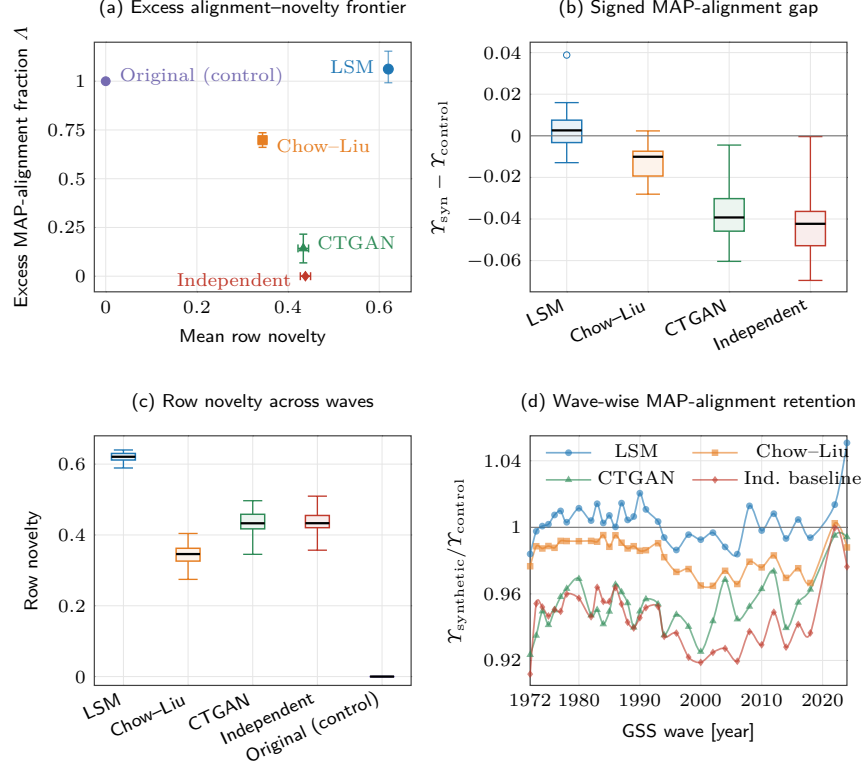


Fig. 1. Four complementary views of the GSS benchmark: (a) the excess MAP-alignment fraction $\Delta_g = (\tilde{\gamma}_g - \tilde{\gamma}_{\text{ind}})/(\tilde{\gamma}_{\text{control}} - \tilde{\gamma}_{\text{ind}})$ against mean row novelty, with nonparametric 95% wave-bootstrap intervals; the independent-marginal baseline is anchored at zero and the original (control) at one; (b) wave-level signed MAP-alignment differences from the original (control); (c) wave-level row novelty, defined as one minus mean nearest-real similarity; and (d) MAP-alignment retention by GSS wave. The label “Chow-Liu” denotes the restricted Chow-Liu hybrid.

compatible with a single joint distribution. MAP alignment remains a well-defined evaluator statistic in that setting, but interpreting d_μ as a metric between underlying generative processes requires the stated compatibility assumption.

The formal development is for finite product spaces. The GSS benchmark uses categorical or discretely encoded variables, while the Gaussian examples use a continuous-density analogue based on an essential supremum. A general treatment of continuous and mixed-type variables, including the effects of discretization, is not established here and is forthcoming.

The benchmark treats GSS records as unweighted observations and does not model survey weights, strata, or primary sampling units. The reported quantities therefore describe conditional structure in the analyzed records rather than population-representative social-science estimands. Missing and nonresponse entries were treated as missing rather than as substantive response categories; the analysis did not statistically impute them. Results may nevertheless depend on the prevalence and pattern of missingness across waves.

Computational cost grows with both the number of coordinates and the number of evaluated records: the evaluator fits one conditional model per variable and scores each record-coordinate pair. Parallelization made the 197–1,273-variable GSS waves tractable, but repeated fitting, extensive tuning, and substantially larger schemas may be costly.

12 Future Work

Immediate extensions are to formalize the continuous and mixed-type theory and to incorporate survey weights and complex-design information into the reference measure and aggregation scheme. Evaluator sensitivity should be assessed through repeated train–test splits, repeated conditional-model fitting, and cross-fitted variants when a separate evaluator-training set is unavailable. Empirical reports should add exact duplicate rates and upper-tail nearest-row similarities, rather than relying only on the mean novelty statistic. Future analyses should also test sensitivity to missingness patterns and variable discretization, together with computational approximations for very high-dimensional data.

13 Data and Code Availability

The GSS data are publicly available from NORC [22]. The LSYNTH implementation and reproducible examples are available in the public repository at <https://github.com/zeroknowledgediscovery/lsynth>. The Python package is distributed through PyPI and can be installed with `pip install lsynth`.

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